THRILL WORLD

3D Immersive Amusement Park Marketing Website

Product Requirements Document (PRD)

Version 1.0 | Ready for Cursor AI Development

Document Info	
Project Name	Thrill World — 3D Immersive Amusement Park Website
Document Type	Product Requirements Document (PRD)
Version	1.0 — Initial Release
Target Stack	Next.js 14 + Three.js / React-Three-Fiber + GSAP ScrollTrigger + Tailwind CSS
Primary Goal	Full 3D immersive scrolling marketing website for an amusement park
Audience	Families, teenagers, thrill-seekers (Age 8–40+)
Delivery Format	Single-page application (SPA) with multi-section scroll

1. Project Overview & Vision

1.1 Executive Summary

Thrill World needs a cutting-edge, visually spectacular marketing website that makes visitors feel the adrenaline and magic of the park before they even arrive. The website must function as a fully immersive 3D experience — not a traditional flat webpage — using scroll-driven animations, parallax depth effects, particle systems, and WebGL-powered 3D scenes.

This PRD defines every section, interaction, animation, component, and technical requirement needed for a Cursor AI agent to build the complete production-ready codebase from scratch.

1.2 Core Design Philosophy

- Dark, cinematic theme with neon accents — think night-time theme park atmosphere
- Scroll = Movement: Every scroll pixel drives an animation or 3D camera movement
- 3D-first: Three.js scenes are the background and content, not decorative add-ons
- Mobile-responsive: Graceful fallback from WebGL to CSS 3D transforms on low-end devices
- Performance-first: 60fps target, lazy-loaded assets, progressive enhancement
- Emotion before information: Users should feel excitement before they read details

1.3 Target Metrics

Metric	Target	Priority
Lighthouse Performance	≥ 85	Critical
First Contentful Paint	< 2.0 seconds	Critical
Time to Interactive	< 3.5 seconds	High
Core Web Vitals (LCP)	< 2.5 seconds	High
Mobile Score	≥ 80	High
Accessibility (a11y)	WCAG 2.1 AA	Medium
Bounce Rate Target	< 45%	Medium
Avg Session Duration	> 2 minutes	Medium

2. Tech Stack & Architecture

2.1 Core Technology Stack

Layer	Technology	Version	Purpose
Framework	Next.js	14.x (App Router)	SSR/SSG, routing, image optimization
3D Engine	Three.js	Latest stable	WebGL 3D scenes and geometries
React 3D	@react-three/fiber	Latest	React wrapper for Three.js
3D Helpers	@react-three/drei	Latest	Camera controls, loaders, effects
Scroll Anim	GSAP + ScrollTrigger	3.x	Scroll-driven animations
Motion	Framer Motion	Latest	UI micro-animations and page transitions
Styling	Tailwind CSS	3.x	Utility-first CSS
3D Post-FX	@react-three/postprocessing	Latest	Bloom, depth-of-field, glitch
Physics (opt)	@react-three/rapier	Latest	Interactive physics elements
Fonts	Google Fonts / Fontsource	—	Custom display typography
Icons	Lucide React	Latest	UI icons
State	Zustand	Latest	Global scroll/section state
Lenis	Lenis Smooth Scroll	Latest	Buttery-smooth scroll behavior

2.2 Project File Structure

```
thrillworld/
```

Recommended folder structure for Cursor:

- src/app/ — Next.js App Router pages
- src/components/ — All React components
- src/components/3d/ — Three.js / R3F scene components
- src/components/sections/ — Full-page scroll sections

- src/components/ui/ — Buttons, cards, modals, nav
- src/hooks/ — Custom hooks (useScroll, useDeviceCapability)
- src/store/ — Zustand state stores
- src/lib/ — Utilities, animation configs, constants
- src/styles/ — Global CSS, Tailwind config
- public/models/ — GLTF/GLB 3D models
- public/textures/ — PBR textures for 3D objects
- public/videos/ — Background video files
- public/sounds/ — Ambient audio (optional)

3. Website Sections — Detailed Specifications

The website is a single long-scroll page divided into 8 main sections. Each section has a full-screen viewport, its own 3D scene or heavy animation, and transitions seamlessly into the next.

3.1 Section 1: Hero / Landing

Hero Section Spec	
Layout	Full viewport (100vw × 100vh), centered content, dark overlay
Background	Animated Three.js particle field — thousands of glowing sparks flying upward, simulating fireworks/confetti from a night show
3D Element	Large rotating 3D Ferris Wheel model in the background, slowly spinning, lit with neon glow shaders
Headline	"Where Every Ride Rewrites Reality" — split-text animated entrance with GSAP
Subheadline	Park tagline with typewriter effect, 1.2s delay
CTA Buttons	"Book Tickets" (neon yellow, primary) + "Explore Park" (ghost/outline, secondary)
Scroll Hint	Animated down-arrow with bounce animation + "Scroll to Explore" label
Entry Anim	Camera starts zoomed into sky, pulls back to reveal full Ferris Wheel — GSAP timeline on page load

Hero: Technical Implementation Notes

- Use @react-three/fiber Canvas as full-screen background (position: fixed, z-index: 0)
- Particle system: BufferGeometry with 5000+ points, custom vertex shader for upward drift
- Ferris Wheel: Load from /public/models/ferris-wheel.glb, add PointLight halos at each car
- Text: Split into individual chars using GSAP SplitText, stagger: 0.04s per character
- Lenis smooth scroll must be initialized at this level for global effect

3.2 Section 2: Attraction Showcase (3D Horizontal Carousel)

Attractions Spec

Scroll Type	Horizontal scroll within vertical scroll using GSAP ScrollTrigger pin
Content	6–8 ride cards, each full-screen width, sliding left as user scrolls down
3D Effect	Each card has a Three.js scene — mini 3D preview of the ride (roller coaster, log flume, etc.)
Card Info	Ride name, thrill level meter (animated), height restriction, duration
Hover FX	Card scales up 1.05×, 3D scene animates (ride starts moving), glow border appears
Thrill Meter	Custom SVG animated bar — fills from 0 to value on card entry
Transitions	Cards enter with slide + fade from right, orchestrated via GSAP timeline
Mobile	Vertical stacked cards with swipe gesture support (touch events)

Attraction Cards — Data Structure

Field	Type	Example Value
id	string	roller-coaster-x
name	string	Thunder Bolt X
tagline	string	0 to 100 km/h in 2 seconds
thrillLevel	number (1–5)	5
minHeight	string	120 cm
duration	string	3 min 20 sec
3dModel	string	/models/roller-coaster.glb
thumbnail	string	/images/rides/thunder-bolt.jpg
category	enum	Thrill | Family | Kids | Water
waitTime	string	25 min (live API)

3.3 Section 3: Immersive 3D Park Map

3D Map Spec	
Type	Interactive overhead 3D map of the park rendered in Three.js
Camera	Isometric-like perspective, OrbitControls for pan/zoom/rotate

Zones	Color-coded zones: Thrill Zone (red), Family Fun (blue), Kids World (yellow), Water World (cyan)
Interactivity	Click on any ride/zone to get popup card with info + "Plan Your Visit" CTA
Animation	Idle: Ferris wheel spins, roller coaster cars move along track, fountains animate
Pathfinding	Highlight shortest path between two selected attractions (optional stretch goal)
Legend	Floating UI overlay on top-right with zone colors and icons
Performance	LOD (Level of Detail) — simpler meshes when camera is far, detailed when close

3.4 Section 4: Park Experience / Video Immersion

Video Section Spec	
Layout	Full-bleed background video, content overlaid
Effect	Scroll-driven clip-path reveal — video starts as small circle, expands to full screen as user scrolls
Video Content	Drone footage / POV ride footage, looping, muted autoplay
Text	Large cinematic text: "3 Million Visitors Can't Be Wrong" — parallax entrance
Stats Row	Animated counters: Total Rides (45+), Annual Visitors (3M+), Years Open (25+), Acres (150+)
Counter Anim	Numbers count up from 0 when section enters viewport (Intersection Observer)
Overlay	Gradient from transparent to dark at bottom for text readability
CTA	"Watch Full Experience" button opens video modal / lightbox

3.5 Section 5: Tickets & Pricing

Tickets Section Spec	
Layout	3-column pricing cards on dark background with star-field particle effect
Cards	Day Pass | Weekend Bundle | Annual Pass — each as a 3D-tilted card

3D Tilt	Cards tilt on mouse-move using CSS perspective transform (vanilla-tilt style)
Highlight	Middle card (Weekend Bundle) is elevated, has glow border, "Most Popular" badge
Pricing	Large price display, feature checklist, animated CTA button
Urgency	"Only 143 tickets left today" — live/simulated countdown
Discount Badge	Rotating ribbon/badge for seasonal discounts
Checkout CTA	Scroll-sticky CTA bar that appears after user passes this section

3.6 Section 6: Guest Stories / Social Proof

Social Proof Spec	
Layout	Masonry-style grid of guest photos + reviews, floating cards at different depths
3D Effect	Cards at varying z-depths, move at different parallax rates on scroll — creates sense of depth/dimension
Reviews	Star rating, quote, guest name, visit date
Photo Wall	User-generated looking photo collage — park moments, family photos, ride selfies
Auto-Scroll	Horizontal strip of review cards auto-scrolls infinitely (CSS marquee style)
Rating Summary	Large "4.9 / 5.0" display with animated star fill and total review count
Social Icons	Instagram, TikTok, Facebook embeds or CTAs to follow

3.7 Section 7: Visit Planning / Info

Plan Your Visit Spec	
Layout	2-column grid: left = info, right = embedded map (Google Maps or 3D map)
Info Cards	Opening Hours, Location, Parking, Food & Dining, Accessibility
Accordion FAQ	Animated expand/collapse FAQ with GSAP height animation
Calendar	Interactive date picker for ticket booking
Weather Widget	Current weather at park location (OpenWeatherMap API)
Directions CTA	"Get Directions" button opens native maps app on mobile

Transport Info	Bus/train/car icons with route info
Entry Animation	Cards slide in from alternating left/right as section enters

3.8 Section 8: Footer

Footer Spec	
Layout	Full-width dark footer with park logo, links grid, newsletter, social
3D Element	Subtle animated skyline silhouette of park rides as footer background
Newsletter	Email signup with animated success state (confetti burst on submit)
Links	About, Careers, Press, Accessibility, Privacy Policy, Terms, Contact
Social Strip	Large social media icons with hover glow effects
Copyright	Year + park name, minimal and clean
Back to Top	Animated rocket/arrow button that smoothly scrolls to hero

4. Animation & Interaction System

4.1 GSAP ScrollTrigger Configuration

All scroll-driven animations must use GSAP ScrollTrigger. Below is the master configuration:

- Smooth scroll: Lenis instance passed to GSAP ScrollTrigger via ScrollTrigger.scrollerProxy()
- Each section registers its own ScrollTrigger with scrub: 1.5 for smooth interpolation
- Use "markers: false" in production, "markers: true" for development
- Pin sections that require sustained scroll (horizontal carousel, video expansion)
- Use timeline.to() chains for coordinated multi-element animations
- Respect prefers-reduced-motion: disable all non-essential animations if set

4.2 Three.js Scene Management

- Single Canvas element with multiple scene "layers" managed by scroll position
- Use @react-three/drei's ScrollControls for 3D scroll within canvas
- Implement scene transitions: fade/dissolve between different 3D environments
- Post-processing pipeline: Bloom → ToneMapping → Vignette for cinematic feel
- Shader materials for glowing neon signs, light trails, and particle effects
- Environment maps (HDRI) for realistic reflections on ride surfaces

4.3 Performance Optimizations

Optimization	Implementation	Impact
Texture Compression	KTX2 format with Basis transcoder	60–80% texture size reduction
Model Optimization	Draco-compressed GLTF files	70% model size reduction
LOD System	Three.js LOD for distance-based detail	40% GPU load reduction
Frustum Culling	Enabled by default (Three.js)	Skip off-screen rendering
Object Pooling	Reuse particle objects instead of creating new	Prevents GC pauses
Code Splitting	Next.js dynamic imports for 3D components	Faster initial load
Image Optimization	Next/Image with WebP + AVIF	50% image size reduction

Preload Critical	Preload hero model + texture on page load	No pop-in on hero
Worker Thread	Offscreen canvas for heavy computation	Main thread stays free

5. UI Components Library

5.1 Design System — Colors & Typography

Token	Value	Usage
--color-primary	#6C2BD9 (Purple)	CTA buttons, headings, brand
--color-accent	#00D4FF (Cyan)	Highlights, glows, borders
--color-energy	#FFD600 (Yellow)	Tickets, warnings, badges
--color-danger	#FF3D71 (Red)	Thrill ratings, urgency
--color-bg-dark	#0A0A1A (Deep Navy)	Page background
--color-bg-card	#12122A	Card backgrounds
--color-text-primary	#FFFFFF	Main text
--color-text-muted	#8080C0	Secondary text, captions
--font-display	Bebas Neue / Russo One	H1, hero text, park name
--font-heading	Sora / DM Sans	H2, H3, section titles
--font-body	Inter / Plus Jakarta Sans	Body copy, descriptions
--font-mono	JetBrains Mono	Ticket codes, counters

5.2 Component Checklist for Cursor

The following components must be built. Each should be in its own file under src/components/:

Navigation Components

- <Navbar /> — Transparent on hero, solid on scroll, mobile hamburger menu
- <MobileMenu /> — Full-screen overlay menu with animated link entrance
- <ProgressBar /> — Thin top bar showing scroll progress through page

3D Scene Components

- <HeroScene /> — Main Three.js canvas with Ferris Wheel + particles
- <RidePreview3D /> — Individual 3D ride model viewer for attraction cards
- <ParkMap3D /> — Interactive isometric 3D park map
- <StarField /> — Animated star/particle background (reusable)
- <NeonText3D /> — 3D extruded text with neon glow material
- <RollerCoaster /> — Animated coaster along procedural track

UI Components

- <Button /> — Primary, Secondary, Ghost variants with hover glow
- <AttractionCard /> — Ride card with 3D tilt, thrill meter, info overlay
- <PricingCard /> — Ticket option card with mouse-tilt 3D effect
- <ReviewCard /> — Guest review with star rating and photo
- <StatCounter /> — Animated number counter on scroll
- <VideoModal /> — Fullscreen video lightbox with blur backdrop
- <ThemeToggle /> — Dark/light mode toggle (dark is default)
- <TicketWidget /> — Sticky floating booking widget on mobile

Section Components

- <HeroSection /> — Section 1: Full hero with 3D background
- <AttractionsSection /> — Section 2: Horizontal scroll attraction cards
- <ParkMapSection /> — Section 3: Interactive 3D map
- <VideoSection /> — Section 4: Immersive video with clip-path scroll
- <TicketsSection /> — Section 5: Pricing cards
- <StoriesSection /> — Section 6: Social proof masonry grid
- <PlanVisitSection /> — Section 7: Info + FAQ + calendar
- <Footer /> — Section 8: Footer with skyline animation

6. Cursor AI — Prompt Strategy

Below are the recommended prompt sequences for building this project with Cursor AI in order:

Step	Cursor Prompt Goal	Key Instructions
1	Project Bootstrap	Init Next.js 14 app router, install all dependencies from Section 2.1, setup Tailwind, configure fonts
2	Design System	Create globals.css with all CSS variables from Section 5.1, setup Tailwind config with custom tokens
3	Layout & Nav	Build Navbar with scroll-aware transparent → solid transition, mobile hamburger with GSAP animation
4	Lenis + GSAP Setup	Initialize Lenis smooth scroll in layout.tsx, proxy to GSAP ScrollTrigger, export reusable hook
5	Hero Scene	Build HeroScene.tsx with Three.js particle system + Ferris Wheel, full-screen canvas, GSAP load animation
6	Hero Section	Build HeroSection.tsx: split-text headline animation, CTA buttons, scroll hint, integrate HeroScene
7	Attraction Cards	Build AttractionCard + AttractionsSection with GSAP horizontal scroll pin + mini 3D ride previews
8	Park Map	Build ParkMap3D with isometric Three.js scene, zone colors, click-to-info interactions, OrbitControls
9	Video Section	Build VideoSection with scroll-driven clip-path circle expand effect + animated stat counters
10	Pricing	Build PricingCard with mouse-tilt effect, highlight animation, urgency countdown, sticky CTA bar

11	Social Proof	Build ReviewCard + StoriesSection with parallax depth grid and infinite auto-scroll strip
12	Plan Visit	Build PlanVisitSection with accordion FAQ, date picker, weather API integration
13	Footer	Build Footer with animated skyline silhouette, newsletter confetti, back-to-top rocket
14	Polish & Perf	Implement LOD, Draco compression, code splitting, prefers-reduced-motion, Lighthouse audit

6.1 Key Cursor Context Comments

Add the following as a comment at the top of every new file for Cursor context:

- // @park: ThillWorld — 3D Immersive Amusement Park Website
- // @section: [Section name]
- // @deps: three, @react-three/fiber, @react-three/drei, gsap, framer-motion
- // @theme: dark (--color-bg-dark: #0A0A1A, --color-primary: #6C2BD9)
- // @perf: target 60fps, use instancing for repeated geometry

7. Content Requirements

7.1 Copy / Text Content

Section	Headline	Body Copy Note
Hero	Where Every Ride Rewrites Reality	One punchy tagline, max 8 words
Attractions	Choose Your Thrill	Brief 1-line description per ride
Park Map	Explore Every Corner	Zone descriptions: 15–20 words each
Video	3 Million Visitors Can't Be Wrong	Stat callouts only, minimal body text
Tickets	Pick Your Adventure	Feature list per tier, 5–7 bullets each
Stories	Real Thrills. Real Smiles.	Short 2-3 sentence guest quotes
Plan Visit	Your Adventure Starts Here	Opening hours, FAQs, directions
Footer	Come Back for More	Tagline only, legal footer copy

7.2 Media Assets Required

- Hero: 1× Ferris Wheel 3D model (GLTF/GLB), night-time park panorama as HDRI
- Attractions: 6–8 ride GLTF models or procedurally generated equivalents
- Park Map: Park layout as a simplified 3D scene (can be procedurally built)
- Video: 1× background loop video (30–60 seconds, < 20MB, H.264/WebM)
- Reviews: 8–12 placeholder guest photos (diverse, cheerful, Creative Commons)
- Icons: SVG icon set for amenities (parking, food, restrooms, first aid, etc.)
- Textures: Neon sign texture maps, park ground texture, night sky HDRI
- Fonts: Bebas Neue, Sora, JetBrains Mono (Google Fonts CDN or Fontsource npm)

8. Mobile & Responsive Behavior

Breakpoint	Width	3D Behavior	Layout Change
Desktop XL	> 1280px	Full WebGL, all effects enabled	Full horizontal layouts
Desktop	1024–1280px	Full WebGL, reduced particles (3000)	Standard layouts
Tablet	768–1024px	WebGL with reduced post-processing	Some 2-col → 1-col
Mobile L	480–768px	CSS 3D transforms, no WebGL	All single column
Mobile S	< 480px	Static illustrations, no animation	Compact layouts

8.1 Mobile-Specific Components

- Sticky bottom "Book Now" CTA bar — appears after scrolling past hero on mobile
- Touch-friendly attraction card swipe (use Swiper.js or custom touch events)
- Park map on mobile: Static image with tap-to-zoom, not full 3D scene
- Hamburger menu: Full-screen overlay with large tap targets (min 44×44px)
- Video section: poster image fallback if autoplay blocked by browser

9. Accessibility (a11y) Requirements

- All interactive elements must have proper aria-label, role, and tabIndex
- Keyboard navigation must work for all interactive elements (cards, map, FAQ)
- Focus indicators must be visible and follow brand styling (neon outline)
- Color contrast ratio: minimum 4.5:1 for body text, 3:1 for large text
- All images must have descriptive alt text
- Videos must have captions or audio description available
- Prefers-reduced-motion: @media (prefers-reduced-motion: reduce) must disable all GSAP animations
- Screen reader: Landmarks (header, main, nav, footer) must be properly implemented
- 3D content must have text alternatives describing what is shown
- Form elements (newsletter, date picker) must have labels and error messages

10. Development Milestones

Phase	Milestone	Duration	Deliverable
Phase 1	Foundation	2 days	Next.js setup, design system, nav, Lenis/GSAP init
Phase 2	Hero + Attractions	3 days	Hero 3D scene, hero section, attraction horizontal scroll
Phase 3	Map + Video	2 days	3D park map with interactions, video clip-path section
Phase 4	Tickets + Social	2 days	Pricing cards with tilt, review masonry, counters
Phase 5	Info + Footer	1 day	FAQ accordion, plan visit info, animated footer
Phase 6	Polish + Perf	2 days	Draco, LOD, mobile fallback, a11y, Lighthouse audit
Total	Production Ready	~12 days	Fully functional, tested, deployable website

Ready to Build — Pass This PRD to Cursor!

Start with Step 1 in Section 6 and work sequentially for best results.